

Resolving the Full Modular Data of a Doubled-Fibonacci String-Net on a Finite Torus by Point-Group Rotation, and an Obstruction to Quasi-One-Dimensional Anyon Exchange

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We report a real-space resolution of the full modular data of a doubled-Fibonacci Levin–Wen string-net, obtained by exact diagonalization on finite torus supercells (2×2 , 2×3 , 3×3), together with a sharply scoped negative result for the dynamical half of the same program — and we keep the classes of evidence strictly separate. (i) The vacuum row of the modular S -matrix, $|S_{0a}| = d_a/D$, is confirmed by *two structurally distinct lattice routes* — an operator route built from a flux basis, and a Verlinde–Casimir route that reads the quantum dimensions from the lattice loop spectrum without using the F or R symbols — the latter reproducing the analytic values at 1.1×10^{-16} and the two agreeing at $< 10^{-9}$, the precision at which the operator-route values are recorded. (ii) The ground-state degeneracy $\text{GSD} = 4$ is a separate check, invariant across 2×2 , 2×3 and 3×3 ; $|S|$ is invariant under 24 sampled $O(4)$ gauge transformations of the ground-state manifold and size-robust to $\leq 2.5 \times 10^{-11}$. (iii) The Wilson-loop algebra *alone* does not fix the chiral sign, and we *measure* that limit rather than assert it: the loop algebra is $\mathbb{C} \oplus M_3(\mathbb{C})$ ($\dim = 10 = 1^2 + 3^2$), its antisymmetric part vanishes identically on the real-symmetric holonomy, and the chiral splitting capacity is zero. This is a concrete instance of a theorem of Li and Mong, who also name the remedy: a point-group rotation of the ground-state manifold, real but *not* symmetric — precisely the property the loop route lacks — available on every $L \times L$ supercell. Carrying it out, we extract the full signed S and T from the lattice on *two* sizes, with no analytic doubled-Fibonacci S or T supplied to the solver, up to the $\mathbb{Z}_2$ relabeling intrinsic to an achiral theory; a validated two-channel error budget with no free parameters accounts on both sizes for the reference error — the distance to that analytic target, formed in a separate scoring step, not an output of the solver. The modular data themselves are not new with this work — Francuz and Dziarmaga obtained them for this model in 2020; new here are the finite-lattice demonstration of the rotation channel, the quantified error budget, and the measured loop-algebra limit as a concrete instance of the Li–Mong bound. Complementing this static certificate, we show that on the one-dimensionally mapped anyonic ladder the Hamiltonian transport of a single mobile Ising anyon around a pinned object cannot realize the non-Abelian braid matrix: the closed-loop holonomy is exactly the identity and $[U(C_1), U(C_2)] = 0$, established through two independent obstructions, three null escape routes, and a machinery-alive control. That obstruction is structural and independent of both the anyon type and the system size, so it constrains the Fibonacci ambition motivating this work and not merely the Ising tracer that exhibits it. Both halves were produced under an adversarial verification harness that, in the course of this work, retracted an attribution of its own — crediting the lattice with the signed data before a channel existed to carry it — and then restored it with proof through the rotation channel. The values were never in question, only their provenance.

I. INTRODUCTION AND FRAMEWORK

A. The original goal

This work began in pursuit of an ambitious target: a real-space braiding simulator. Concretely, to carry two non-Abelian Fibonacci anyons (τ) around one another *in real space* and to measure the non-Abelian Berry/monodromy matrix they accumulate, whose target value is

$$M_{\tau\tau} = -1/\varphi^2 = -0.382, \quad (1)$$

with $\varphi = (1 + \sqrt{5})/2$ the golden ratio. This quantity is a characteristic monodromy diagnostic of Fibonacci

statistics; universality rests on the matrix-valued braid action, of which this scalar is one invariant. Obtaining it *Hamiltonian-dynamically*, through genuine quasiparticle transport rather than as a digital gate sequence on a qubit chip, would be categorically distinct from the recent digital demonstrations of Fibonacci anyons on quantum processors [1], which realize the statistics as gate circuits with no static Hamiltonian and no transport dynamics.

The goal splits cleanly into a static and a dynamic half, and the line between them is where this paper sits. The static half is delivered: we extract, from a genuine finite-lattice computation, the basis-invariant modular content of the doubled-Fibonacci Levin–Wen string-net [2]. The dynamic half is a verified no-go on this model: the transport is structurally trivial. Both outcomes were anticipated as publishable before the decisive test was run,

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which is why we report them together.

B. The narrow ridge, told plainly

The approach to the dynamic half was sound, and stating that plainly is part of the contribution. Fibonacci dynamics live in a small fusion space that is tractable locally via anyonic matrix product states [6], at a cost scaling as χ^3 rather than exponentially, so no large-scale computing resources were required. A tracer-first ladder of validations was passed against known answers, and both target braid matrices — the Fibonacci monodromy of Eq. (1) and the Ising tracer $e^{-i\pi/4}\sigma_x$ — were built exactly from the framework’s own F and R symbols.

That left one decisive question: does the transport actually realize the braiding? The clean test was an Ising-Berry gate — transport a free-fermion Ising anyon, whose braid answer is known in closed form, and check whether the measured matrix hits it. Either branch was publishable: a hit would carry the method through to Fibonacci, a miss would fix the boundary honestly. The single real hazard was asymmetric: the hopping Hamiltonian is given only diagrammatically in the source papers [6, 7], so it had to be reconstructed from the originals rather than guessed, because a mis-built anyon returns a plausible-looking but meaningless number.

The decisive test resolved to the no-go of Sec. IV.

C. Why an Ising test settles a Fibonacci question

Section IA motivates Fibonacci; Sec. IV tests Ising. The bridge is that *the obstruction is structural, and independent of the anyon type and of the system size*. The two mechanisms established in Sec. IV are gauge-removability of the braid R -phase on a simply connected graph, and superselection of the total fusion charge. Neither invokes the specific fusion category: the first is a statement about closed walks on a graph, the second about the block structure of every local operator in the model. Neither depends on the chain length N ; we verify the flux statement explicitly on $N = 3, 5, 7$.

Ising is therefore not a substitute target but the *decisive* test: its braid answer is known in closed form, so a null result cannot be attributed to an unknown target or to a mis-tuned parameter. The no-go consequently constrains the Fibonacci ambition that motivates this work, and not merely the Ising tracer that exhibits it.

D. The harness

Both halves were produced under an adversarial verification harness: every positive claim must survive a battery of locks, each lock must carry a positive control that makes it fail on a known-nontrivial input, and any claim that cannot be reproduced from the deposited artifacts is

withdrawn. The harness is not decoration. In the course of this work it corrected an attribution of its own, twice. An earlier internal headline had credited the lattice with the *full, signed* modular S without proof; an independent recomputation withdrew that credit, showing that the raw signed matrix of the operator route is not symmetric, is not gauge-equivalent to the analytic S , and fails the modular consistency condition at $(ST)^3 - S^2 = 1.27$ — facts that remain true of that route. A follow-up then over-corrected, declaring the lattice unable to carry the sign at all; a point-group rotation of the ground-state manifold, until then untested, restored the attribution with proof. Section III traces both corrections and the channel that resolved them. What moved was never the values, only their provenance — a discipline we regard as the honest core of the paper rather than a damage report.

E. Model

We work with the doubled-Fibonacci Levin–Wen string-net [2] on an $L \times L$ torus. The plaquette operators B_p are built from the (real) Fibonacci F -symbols; the ground space is the joint $+1$ eigenspace of the B_p . The anyon content is the Drinfeld center, four sectors $a \in \{(\mathbb{1}, \mathbb{1}), (\tau, \mathbb{1}), (\mathbb{1}, \bar{\tau}), (\tau, \bar{\tau})\}$ with quantum dimensions $\{1, \varphi, \varphi, \varphi^2\}$ and total quantum dimension $D = \sqrt{\sum_a d_a^2} = 1 + \varphi^2 = 3.618\dots$, so that $D^2 = 13.09\dots$. Exact diagonalization is performed at 2×2 , 2×3 and 3×3 , of Hilbert-space dimension 175, 2250 and 106250 respectively.

We stress a scope point up front, because it bounds everything below: a second lattice *size* beyond 3×3 is not reachable by exact diagonalization. A 4×4 state requires ~ 13 GB per vector and 5×5 about 1.4 PB per vector. Where a claim rests on 3×3 alone we say so.

II. LATTICE-EMERGENT MODULAR CONTENT

The central discipline of this section is that the classes of evidence are *not* interchangeable, and we label each one: (a) the vacuum row, fixed by two structurally distinct lattice routes; (b) the topological spins, which the loop overlaps constrain only algebraically; and (c) the ground-state degeneracy, a separate lock. The full signed data rest on a fourth class, the point-group rotation channel, which we develop in Sec. III. Bundling these into a single “independent routes confirm our result” would be an overclaim; keeping them apart is what let the harness catch one earlier in this work.

A. Class (a): the vacuum row, by two distinct lattice routes

Only for the vacuum row of S do two structurally distinct *lattice* routes exist, and they agree.

The *operator route* builds the commutant of $\{B_p\}$, forms the K -even and K -odd ground-space actions of a Wilson-loop operator W along a noncontractible cycle, diagonalizes them jointly to obtain a flux basis, and reads $S = U_i^\dagger U_j$ as the overlap of the flux bases of the two cycles.

The *Verlinde–Casimir route* shares none of those code paths. It resolves a four-sector basis from a single hermitian lattice Casimir $H_W = 0.7 M_i - 0.3 M_j$, where $M_g = V^\dagger W_g V$ is the ground-space action of the cycle loop; because the two cycle loops fail to commute at finite size, $\|[M_i, M_j]\|_{\max} = 5.6 \times 10^{-2}$ (largest absolute entry), this noncommutativity lifts the per-cycle degeneracy. It then reads $S_{0a} = d_a/D$ directly from the lattice loop spectrum via the Verlinde relation [8], rather than from any basis overlap. Its extraction inputs are only $\{V_{\text{gs}}, W_{\text{cycle } i}, W_{\text{cycle } j}\}$; no F , no R , and no analytic S enters.

We separate the two comparisons, because they are recorded at different precision. The Verlinde–Casimir route, independent of analytic F/R symbols and of the target modular matrix, reproduces the analytic values $\{0.2764, 0.4472, 0.4472, 0.7236\}$ at 1.1×10^{-16} . The two *routes* agree with each other at $< 10^{-9}$ — that bound, and not the 10^{-16} figure, is what the operator-route data support, since those values are recorded to four digits. Conflating the two would overstate the route-to-route agreement by seven orders of magnitude. This is the only claim of two structurally distinct lattice routes made anywhere in this paper, and it applies to the vacuum row alone.

B. Class (b): topological spins — algebraic from the loops, lattice-determined by rotation

The topological spins are $T = \text{diag}(1, e^{4\pi i/5}, e^{-4\pi i/5}, 1)$, i.e. spins $\{0, 0, \pm 4\pi/5\}$. From the loop overlaps *alone* they are fixed only noncircularly and algebraically — no analytic T is inserted — by two consistency paths applied to the lattice S ; the direct lattice *measurement* of T comes not from the loops but from the point-group rotation channel of Sec. III. The two consistency paths are:

- (b1) a *Verlinde fixed point*: solving $(ST)^3 = S^2$ from random grid starts (residual 3.98×10^{-11}). This is a consistency check, and by itself it does not pin T uniquely, since S leaves Galois orbits open.
- (b2) a *Gauss-sum/anomaly* argument [9, 11]: $p_+ = \sum_a d_a^2 \theta_a = D$ at $c = 0$ pins the solution uniquely without invoking $(ST)^3$, and rejects the order-3 alternative ($5.236 \neq D$). The Galois-conjugate so-

lution is excluded not by this argument but by the positivity of the extracted quantum dimensions ($d_\tau = \varphi > 0$ rather than $-1/\varphi$), i.e. by unitarity [14]: Galois conjugation preserves the order of T , so a congruence of level 5 cannot separate members of one Galois orbit [12, 13].

Both paths take the lattice S as input and reason about it algebraically. Two facts about the loops explain why this is as far as they reach, and they set up the channel that reaches further. Real-symmetric lattice loops are phase-free, so a Wilson-loop overlap cannot carry a spin; and the geometric Dehn twist available on a 3×3 torus has order 3, whereas the theory’s T has order 5, so that twist cannot generate T either — a directly geometric order-5 twist would require $L \equiv 0 \pmod{5}$. Neither obstacle touches the point-group rotation ST , a *different* modular element whose order is 3 intrinsically and on every size; the order-3 limitation that blocks the Dehn twist does not apply to it. That rotation is the channel of Sec. III.

C. Class (c): ground-state degeneracy

GSD = 4 is a separate check and a separate class of evidence. The degeneracy and the gap are invariant across all three lattice sizes: four ground states, clean gap 1.0.

The ground-state *energy* is not, and we state the scope rather than let a reader infer one. E_0 is *extensive*: it counts one unit per plaquette, so $E_0 = -4$ on 2×2 and $E_0 = -9$ on 3×3 . It is therefore the degeneracy and the gap — not the energy — that carry the invariance, and quoting a single E_0 for the model would be meaningless.

D. Soundness of the $|S|$ extraction

Two properties support the $|S|$ extraction. They are properties of $|S|$; they do not extend to classes (b) or (c), each of which stands on its own evidence.

Gauge invariance. $|S|$ is invariant under $V \rightarrow V \cdot Q$ over 24 sampled $O(4)$ gauge transformations of the ground-state manifold, with a worst deviation of 2.4×10^{-15} on 2×2 and 5.6×10^{-15} on 2×3 . That $O(4)$ is the correct residual gauge group is itself a lattice fact: the F -symbols are real, hence the Hamiltonian is real and time-reversal invariant, hence the K -odd part vanishes (6.9×10^{-17}) and the ground states may be chosen real. The corresponding $U(4)$ test *fails* at 0.603 and thereby acts as a discriminator: a test that passes for every group would be measuring nothing.

Size robustness. $|S|$ agrees across 2×2 , 2×3 and 3×3 to $\leq 2.5 \times 10^{-11}$, and the $|S|$ multiset $\{0.2764, 0.4472, 0.7236\}$ reproduces the doubled-Fibonacci values.

The string operator underlying the extraction is genuine in a checked sense: it commutes with every plaquette operator ($\leq 1.7 \times 10^{-16}$), it has support outside

the ground space, and it acts nontrivially on the ground-state manifold. Its diagonal Wilson line reproduces the doubled-Fibonacci eigenvalues $\{\varphi^2, -1, -1, 1/\varphi^2\}$ exactly, with error 0.0. The entry $+1/\varphi^2$ is a static lattice quantity; it must not be confused with the dynamical target $M_{\tau\tau} = -1/\varphi^2$ of Eq. (1), which carries the opposite sign, belongs to the other half of the program, and was never measured (Sec. IV). We call the operator *chirality-discriminating* (equivalently, *R*-phase-sensitive) rather than chiral: the theory is achiral, $c = 0$.

Honesty requires one further statement here. The $|S|$ /quantum-dimension extraction itself is *single-route* on 3×3 : it carries no two-route redundancy at that size, and 5×5 is infeasible by exact diagonalization. What supports it is the genuine out-of-ground-state, chirality-discriminating string, not a second independent measurement.

E. Consistency locks

The categorical data underlying B_p satisfy the hexagon equation to 2.4×10^{-16} and the pentagon equation with zero violations, and the Levin–Wen hermiticity, fusion, projector and commutation properties hold at $\sim 10^{-16}$. These locks are not tautological: a deliberately naive evaluation of the same expression, $FRF - RFR$, returns 1.18 rather than zero, so the lock discriminates.

III. THE HONEST CUT: WHAT THE LOOPS CANNOT GIVE, AND HOW THE LATTICE GIVES IT

A. The loop route, measured: an instance of the Li–Mong bound

The operator route of Sec. II A fixes the magnitudes $|S|$ but not the chiral sign, and we measure that limit rather than assert it. Two facts establish it.

First, the raw signed matrix that route produces is not a valid modular S : it is not symmetric (symmetry error 1.46, whereas a valid S must be symmetric), it is not gauge-equivalent to the analytic S (best over 24 permutations and 16 sign choices: 1.08, against a requirement of $\sim 10^{-2}$), it fails modular consistency at $(ST)^3 - S^2 = 1.27$, and the Verlinde relation applied to it returns negative, noninteger fusion multiplicities; the corresponding canonical- S lock returns FALSE. These are honest results of the project’s own audit, and they remain true of that route. A selector-backed variant of it did reach a 1.16×10^{-10} match to the analytic S/T — but only by *supplying* the sign from an $SL(2, \mathbb{Z})$ solvability argument rather than measuring it; we report that number only as a property of the superseded route.

Second, the reason is structural, and we can exhibit it. On 3×3 the cycle holonomies M_i, M_j are exactly real-symmetric, and their antisymmetric parts vanish identi-

cally ($\|i(M - M^T)/2\|_F \approx 1.2 \times 10^{-16}$ and 7.5×10^{-17}). The loop algebra $\langle M_i, M_j \rangle$ has dimension $10 = 1^2 + 3^2$, i.e. $\mathbb{C} \oplus M_3(\mathbb{C})$: with only $\{\mathbb{1}, \tau \times \bar{\tau}\}$ Wilson loops available, the four anyon sectors cannot be separated into four one-dimensional blocks. Each real-symmetric loop resolves only three distinct levels, the pair $(\tau, \mathbb{1})/(\mathbb{1}, \bar{\tau})$ remaining degenerate, and the chiral splitting capacity of that pair is 0.0. A single hermitian Casimir does resolve the four-sector *basis*, but a basis is not a signed S .

This is not a shortfall of our implementation; it is a concrete instance of a theorem. Li and Mong [3] prove that Wilson-loop and minimum-entropy-state overlaps determine only the absolute-value structure of the modular data, leaving the individual spins θ_a inaccessible. Our measured loop algebra realizes that bound on a finite lattice. The remedy is theirs too, and the rest of this section takes it up.

One internal quantity must be named to prevent a spurious contradiction. A separate “ e –15 match” of $|S|$ row multisets used *different* row and column permutations; such a match constrains the multiset of absolute values only, is not a representation-preserving anyon relabeling, and is a different object from the 1.16×10^{-10} above. The two must not be conflated — this “value correct, scope wrong” distinction is one the harness enforces throughout.

B. The rotation channel

The remedy Li and Mong name is a point-group rotation of the ground-state manifold, available on every $L \times L$ supercell. On our lattice it is the 120° rotation R about a plaquette center; realized on the ground space as $M_R = V_{\text{gs}}^T R V_{\text{gs}}$, it is a genuine symmetry — orthogonal to 2.96×10^{-15} on 3×3 and 4.41×10^{-15} on 2×2 , of order three — and it does *not* lie in the loop algebra: adjoining it raises the algebra dimension from 10 to 16.

Why the loops missed it is the most instructive line of this paper. A real-symmetric operator has real eigenvalues and cannot carry a phase; the Wilson-loop holonomies are exactly such operators, which is why they fix only $|S|$. A rotation is real but *not* symmetric, so it can. The earlier statement that “the 3×3 geometry is real-symmetric and therefore phase-free” was true of the Wilson-loop observable, not of the lattice — the lattice carries other real operators that are not symmetric, and M_R is one.

Because M_R commutes with neither M_i nor M_j , it lifts the degeneracy the loops leave. From the measured pair (M_{Wilson}, M_R) we solve for the modular data, the rotation supplying the phase and the loop spectrum the magnitudes. On 3×3 the solver returns four solutions per computed rotation center, with residuals 2.9×10^{-10} to 4.8×10^{-10} and the correct sign structure on every one, yielding $T = \{e^{\mp 4\pi i/5}, 1, 1\}$.

Those centers are not independent repetitions, and we say so rather than let the count suggest otherwise. On

an $L \times L$ torus the nine possible centers generate only three distinct edge permutations; centers sharing a permutation perform the same computation and their projected rotations therefore agree bit for bit, while across all nine the projected rotation agrees to 1.4×10^{-15} , the floating-point floor. Two of the three centers used here fall in the same class, so the twelve solutions are eight distinct ones. We deposit one rotation matrix per class and a script that checks this from the record. Physically the content is that lattice translations act trivially on the ground space, as topological order on the torus requires; a center-dependent modular datum would have been the surprise. Running further centers is therefore a consistency check on the optimizer, not an independent repetition — the independent repetition in this work is the second lattice size.

On 2×2 the solver returns 9 solutions across three centers, residuals 6.4×10^{-10} to 1.8×10^{-9} , again with correct sign structure; the extracted T spectra agree across all three centers at the level of the extraction residuals themselves, up to the intrinsic $\mathbb{Z}_2$ relabeling ($\tau \leftrightarrow \bar{\tau}$) — verifiable from `p6_extraction_2x2.json`. The full signed S and T are thereby resolved from the lattice on two sizes. These are two *sizes* of one lattice route, not two routes: the signed data rest on the single rotation channel, and no second structurally distinct lattice route is claimed for them.

The extractor merits a precise description, since it is released with the paper. It solves a nonlinear least-squares problem whose residual encodes generic modular-tensor-category axioms — S symmetric, a real nonnegative vacuum row, $|\theta_a| = 1$, fusion-character consistency, S^2 equal to charge conjugation, and $(ST)^3 = \zeta S^2$ with $|\zeta| = 1$ (the central-charge phase, left free). These axioms *define* the object sought and are applied identically to every input. No analytic S or T , no F or R symbol, and no solvability selector is fed in; a poisoning test confirms the theory helpers are never reached. Verlinde integrality and S -unitarity are applied as output gates after the solve, and an independent referee harness rechecks the result against separately held thresholds. The chiral sign enters through the measured rotation M_R ; the modular axioms only sieve out nonmodular branches. This is the structural difference from the superseded route, where the sign was *supplied* by a solvability selector rather than *measured*. That the rotation must equal $V(S \cdot T^*)V^\dagger$ up to a permutation is itself a consequence of the Li–Mong analysis, not an assumption: the only rotation action consistent with the fusion data is a permutation preserving both quantum dimension and spin, with Wilson phases drawn from the Abelian subcategory; for doubled Fibonacci that subcategory is trivial and the permutation is forced to the identity, so this form is the unique one allowed.

C. Why the extraction is unique for doubled Fibonacci

That the extraction returns a unique answer for doubled Fibonacci is luck rather than merit — and it is again Li and Mong’s theorem, not our inference, that says so. They prove that the number of fusion phases left free by the loop-plus-rotation data equals the number of *self-dual* Abelian anyons. Doubled Fibonacci has quantum dimensions $\{1, \varphi, \varphi, \varphi^2\}$, so its only Abelian anyon is the vacuum—which is self-dual—giving a count of 1: the single fusion phase is trivial, and S and T are fixed. Fibonacci is simply the most favorable case of their theorem.

The freedom is real when $|\mathcal{A}| > 1$. A related but distinct degeneracy shows up empirically in our own inverse problem: on the fully Abelian toric code the extractor returns *two* solution classes rather than one (residuals 1.0×10^{-15} and 1.3×10^{-15}), while on doubled Fibonacci it returns a single class. This nonuniqueness is a degeneracy of *our* data (one loop spectrum plus M_R) rather than an instance of Li and Mong’s fusion-phase freedom, and it arises because those data do not separate the two classes; its absence on Fibonacci is the strongest evidence that the doubled-Fibonacci extraction is well posed.

D. Which category the data selects

Among the unitary modular tensor categories of rank 4 classified in Ref. [10], exactly one is compatible with the extracted data: $Z(\text{Fib}) = \text{Fib} \boxtimes \text{Fib}$. The quantum dimensions alone do not suffice — $\text{Fib} \boxtimes \text{Fib}$ and $\text{Fib} \boxtimes \text{Fib}$ share them exactly — and are separated only by the topological spins, whose multiset $\{0, 0, \pm 4\pi/5\}$ is reproduced by the achiral combination alone. The residual $\mathbb{Z}_2$ is an automorphism of that single category (exchange of the two chiral factors), not a competing candidate. All excluded competitors differ by $O(1)$, so the conclusion is independent of the error budget.

E. The error budget

The quantity budgeted here is the reference error: the distance between the extracted data and the analytic doubled-Fibonacci target, formed in a separate scoring step and not an output of the solver. It is not a single number but a two-channel budget, and validating it across both lattice sizes with no free parameters is a result in its own right. This residual follows

$$\text{residual} \approx 2.081 \varepsilon_{\text{frame}} + k s^2, \quad (2)$$

where $s = \varepsilon_{\text{ideal}}/(3.2668 \times 10^{-3})$ is the loop ideality relative to its 3×3 value, with two independently measured channels. The eigenvector (frame) channel is linear, its transfer coefficient 2.081 constant to 0.02% over four and

a half decades. The eigenvalue channel is quadratic, with k between 4.3×10^{-12} and 6.3×10^{-11} . Both coefficients were deposited before the 2×2 run.

The first-order projection is demonstrated directly: the fusion-character residual is invariant under 300 random ground-space frame choices (relative scatter 6.2×10^{-9}), because the dominant lattice error — a fusion-character deviation of 3.3×10^{-3} — takes the same value on the degenerate pair to within 4.8×10^{-11} and so projects out at first order.

What makes this evidence rather than a fit is that the incomplete version fails. A pre-registered single-channel formula, keeping only the frame term, predicted the 2×2 residual at 4.4×10^{-15} ; the measured value is 6.4×10^{-10} to 1.8×10^{-9} , six orders of magnitude larger. Only the full two-channel law is consistent with both sizes: over all points its conservative prediction interval, 2.0×10^{-10} to 2.9×10^{-9} , contains the entire measured interval, the worst-case edge ratio being 9.2. The dominant channel even switches between sizes — the frame term on 3×3 , from LOBPCG ground states, the quadratic term on 2×2 , from dense diagonalization — and a counterintuitive consequence holds: although the 2×2 loops are $6.8 \times$ less ideal, the extraction degrades only $3.1 \times$, because on 2×2 the frame is essentially exact. The scale s is set by the deposited loop ideality and is never a free knob.

F. What remains a convention: splits versus labeling

One genuine limit survives, and it is a property of the theory, not of the method. The doubled-Fibonacci theory is achiral ($c = 0$), and complex conjugation K is an exact antiunitary symmetry of the real lattice model ($\bar{T} = PTP$ for a permutation P). The rotation channel therefore *separates* the chiral pair but does not *label* it: which of $\tau, \bar{\tau}$ is called which remains a convention, for this method and for any other, Francuz–Dziarmaga included. Empirically the run lands on either convention indifferently — the conjugate match is mixed, 2 of the 8 distinct solutions on 3×3 and 5 of 9 on 2×2 — which is the prediction, not a contradiction. This $\mathbb{Z}_2$ relabeling is a different thing from the fusion-phase freedom of Sec. III C: the latter is trivial for doubled Fibonacci, the former is intrinsic to any achiral theory.

G. What is not used as a certificate

We do not use a tube-algebra cross-check [15] as an independent certificate, although it would be natural to reach for. For a doubled theory it is circular: $Z(\mathcal{C}) = \mathcal{C} \boxtimes \bar{\mathcal{C}}$ is a theorem, so any theory-side center computation necessarily factorizes into $S_{\text{Fib}} \otimes \bar{S}_{\text{Fib}}$ (verified here at 1.1×10^{-16}). “Two theory methods agree” is then guaranteed by the theorem and checks nothing. We cite [15] as a methods and context reference only.

IV. A NO-GO FOR QUASI-ONE-DIMENSIONAL ANYON EXCHANGE

A. Scope

We report a bounded negative result. On the one-dimensionally mapped anyonic ladder of Singh, Pfeifer, Vidal and Brennen [6, 7], the dynamical, Hamiltonian-driven transport of a single mobile σ anyon around a pinned object *cannot* realize the non-Abelian Ising braid matrix. The closed-loop holonomy is exactly the identity, and the commutator of two independent loop holonomies vanishes,

$$[U(C_1), U(C_2)] = 0. \quad (3)$$

We establish this through two independent obstructions, test three escape routes (all null, with no tuning), and — the point that makes a null number trustworthy — verify with a machinery-alive control that the vanishing is physics and not a dead numerical pipeline.

This is a statement about *this model*. It is not a claim that non-Abelian braiding is impossible in general, nor that a two-dimensional geometry cannot realize it. The static certificate of Sec. II is untouched by it.

B. The result

Let G be the (position $\times$ leg) graph of the mapped ladder, with the mobile σ living on a vertex. The construction has four load-bearing hypotheses: (i) G is simply connected, so that all cycles are contractible; (ii) the braid R -phase resides purely in the bond amplitudes, $x = R_{\mathbb{1}}^{\sigma\sigma} J$ and $y = R_{\psi}^{\sigma\sigma} J$, identically on both legs; (iii) a single mobile anyon traverses a closed loop; and (iv) the encircled object is positionless, living in the fusion space rather than on G .

Under these hypotheses the argument is short and exact: total charge is fixed, so each fusion sector is one-dimensional; the Berry connection is therefore Abelian $U(1)$ per sector; the net R -flux vanishes on every contractible cycle; the R -phase is pure gauge on G ; hence $U(C) = \mathbb{1}$ exactly, and Eq. (3) follows.

Numerically: plaquette R -flux = 0; gauge-reconstruction residual 1.7×10^{-16} ; spectrum invariance under the gauge transform 2.1×10^{-15} ; the full H_1 cycle-basis maximal flux = 0 on $N = 3, 5, 7$; and a dense exact-diagonalization adiabatic transport giving Eq. (3) with operator and Frobenius norm 0 and $U(C_1) = U(C_2) = \mathbb{1}$.

The same statement is deposited independently at the *structural* level: building each holonomy directly from the exact R -symbols as $U(C) = \text{diag}(e^{i \text{net}} \mathbb{1}, e^{i \text{net}_\psi})$ gives $\|U(C_1) - \mathbb{1}\|_F = \|U(C_2) - \mathbb{1}\|_F = 0$ and $\|[U(C_1), U(C_2)]\|_F = \|[U(C_1), U(C_2)]\|_{\text{op}} = 0$. We label the two levels separately and do not merge them into one “verified” number: the deposited structural route and the dense-ED adiabatic transport are two levels of

TABLE I. Load-bearing quantities, with evidence class and scope. Class (a) is the vacuum row, fixed by two distinct lattice routes; (b) the loop route and the consistency checks it supports; (c) the ground-state degeneracy, a separate lock; (d) the point-group rotation channel that resolves the full signed S and T . The (b) rows record what the loop route measures and, by measuring it, cannot reach; the (d) rows record what resolves it. Values are reproduced from the deposited artifacts (see Data and Code Availability).

Quantity	Value	Class	Scope
Vacuum row $ S_{0a} = d_a/D$	$\{0.2764, 0.4472, 0.4472, 0.7236\}$	(a) two lattice routes	3×3
Verlinde–Casimir vs. theory	1.1×10^{-16}	(a) one route, no analytic F/R	3×3
route to route	$< 10^{-9}$	(a) operator $\wedge$ Verlinde–Casimir	3×3
S_{00}	0.2763932 , error 1.7×10^{-15} ; $= 1/D$	(a) lattice	2×2
GSD, gap	4, 1.0	(c) separate lock	all three sizes
E_0 (extensive, <i>not</i> inv.)	-4 (2×2), -9 (3×3)	(c) context	size-specific
Gauge invariance of $ S $	2.4×10^{-15} (2×2), 5.6×10^{-15} (2×3)	soundness	$24 \times O(4)$
$U(4)$ control	fails at 0.603	discriminator	—
Size robustness of $ S $	$\leq 2.5 \times 10^{-11}$	soundness	$175/2250/106250$
Hexagon / pentagon / Levin–Wen	2.4×10^{-16} / 0 / $\sim 10^{-16}$	consistency locks	naive: 1.18
Topological spins T	$\{0, 0, \pm 4\pi/5\}$; Verlinde resid. 3.98×10^{-11}	(b) consistency; measured in (d)	lattice S
Loop algebra $\langle M_i, M_j \rangle$	$\dim = 10 = 1^2 + 3^2$	(b) loop route, measured limit	3×3
Antisym. split; chiral capacity	1.2×10^{-16} , 7.5×10^{-17} ; 0.0	(b) loop route, measured	3×3
Raw signed S , loop route	sym 1.46; gauge 1.08; $(ST)^3 - S^2$ 1.27; lock FALSE	(b) loop route fails, measured	3×3
Selector-backed old route	1.16×10^{-10} vs. theory	superseded: sign supplied	not lattice-borne
C_3 orthogonality dev.	2.96×10^{-15} , 4.41×10^{-15}	(d) genuine symmetry	$3 \times 3/2 \times 2$
Algebra jump with C_3	$10 \rightarrow 16$ ($M_R \notin$ loops)	(d) rotation $\neq$ loops	3×3
Signed S, T extraction	12 sol.; res. $2.9\text{--}4.8 \times 10^{-10}$; $T = \{e^{\mp 4\pi i/5}, 1, 1\}$	(d) rotation channel	3×3
Signed S, T extraction	9 sol.; res. $6.4 \times 10^{-10}\text{--}1.8 \times 10^{-9}$	(d) rotation channel	2×2
$\mathbb{Z}_2$ convention (conj. match)	mixed: 2/8, 5/9	(d) intrinsic to achiral theory	$3 \times 3/2 \times 2$
Frame transfer (eigenvector ch.)	2.081, linear, 0.02% over 4.5 dec.	error budget	model
k (eigenvalue channel)	$4.3 \times 10^{-12}\text{--}6.3 \times 10^{-11}$, quadratic	error budget	model
Loop ideality (input scale)	$2.2 \times 10^{-2} \rightarrow 3.3 \times 10^{-3}$	deposited, not a knob	$2 \times 2/3 \times 3$
Two-channel prediction	$2.0 \times 10^{-10}\text{--}2.9 \times 10^{-9} \supseteq$ meas. $6.4 \times 10^{-10}\text{--}1.8 \times 10^{-9}$	conservative	both

evidence for one statement, the latter being the independent confirmation of the former.

Two positive controls keep the null honest. The holonomy builder is live: a loop with *nonzero* net winding returns $\|U_W - \mathbb{1}\|_F = 2 \neq 0$. The commutator builder is live: the same norm code path, fed an F -dressed off-diagonal pair, returns $\sqrt{2} \neq 0$, while the $F = \mathbb{1}$ antifudge collapses it to exactly 0. We note explicitly what we did *not* use as a control: two winding loops over the same bond-phase pair are powers of one Abelian generator, so their commutator vanishes trivially and would prove nothing. An early version of the control was exactly that vacuous, and was discarded.

C. Two independent obstructions

Topological. A one-dimensional strip cannot enclose a point defect: any closed walk crosses each bond a net zero number of times, so the R -phase cancels and is gauge removable. Here $\text{net}_{\mathbb{1}} = \text{net}_{\psi} = 0.0$. This is independently consistent with the graph-braid-group classification of anyons on quantum wire networks [16, 17], in which a network that is not triconnected does not support planar encircling braiding.

Algebraic (superselection). The total Ising fusion charge $\{\mathbb{1}, \psi\}$ is an exact superselection sector: both hop-

ping and pinning have zero matrix element between the $\mathbb{1}$ and ψ blocks. The non-Abelian generator $2\sigma_y$ requires an $\mathbb{1} \leftrightarrow \psi$ rotation — the F -move — and no local or spatial operation generates it. Over 10^4 random transports diagonal in $\{\mathbb{1}, \psi\}$ the commutator vanishes: the deposited value is exactly 0, as it must be, since operators diagonal in a common basis commute by construction. An independent reimplement of the *same* test reproduces this to its floating-point floor (1.57×10^{-16} , a one-ulp artifact of the complex matrix multiply, not a physical residue, and not a second independent check).

The *mechanism* is confirmed by a separate discriminator: the non-Abelian commutator exists only through the F -move. With the true (Hadamard) F one finds $\|[s_1, s_2]\|_F = \sqrt{2}$; setting $F = \mathbb{1}$ forces it to exactly 0.

The two obstructions are the same cause seen two ways, one topological and one algebraic, and so constitute two independent proof paths for a single no-go. The step “flux = 0 $\Rightarrow$ trivial holonomy” holds only for Abelian connections; it is airtight here because each fixed-charge sector is scalar per channel.

TABLE II. The three escape routes, each relaxing a load-bearing hypothesis of Sec. IV B. All are null; none required tuning or a convention flip.

Escape	Relaxed	Outcome
Two- σ exchange	(iii), (iv)	Exchange is diagonal/Abelian; [B_{12}, B_{23}] = 0 in both norms. Cause: superselection.
Periodic ring	(i)	A winding phase survives but is unchanged under bond shift: an Aharonov–Bohm hole flux, not defect encircling; and Abelian.
Direct B operator	bypasses space	Oracle artifact: $\ [B_{1m}, B_{2m}]\ _F = \sqrt{2}$ for <i>every</i> identification, i.e. zero emergent content.

D. Escape routes: live-tested, anti-fudge-gated, all null

Table II summarizes. The first relaxes hypotheses (iii) and (iv): an $N = 4$ dense-ED two- σ exchange, on blocks of dimension 48 each, with the graph fully connected so that a path exists. The exchange is diagonal and Abelian ($\varphi_\psi - \varphi_{\mathbb{1}} \approx \pi$) and the braid generators commute in both norms; the cause is again superselection. The second relaxes (i): on an $L = 6$ ring a winding phase does survive and is not gauge removable, but shifting the bond leaves it unchanged, which identifies it as an Aharonov–Bohm flux through the lattice hole rather than defect encircling — and it is Abelian in any case. The third bypasses space entirely, building B from F and R on the fusion space; it returns the same nonzero value for *every* identification, which marks it as an oracle rather than a demonstration, and the corresponding lock correctly labels it as such.

Each route carried its own controls: machinery-alive, gauge invariance, and an oracle-versus-simulation separation for the first; a braid-free null and an injected $\pi/2$ read back for the second; and the oracle lock for the third.

E. Machinery-alive control

The identical singular-value/Wilczek–Zee transport pipeline [18], applied to a synthetic control carrying a genuine π rotation (a 2π -solid-angle loop), returns $U_{\text{control}} = -\mathbb{1}$, with

$$\|U_{\text{control}} - \mathbb{1}\|_F = 2\sqrt{2} \approx 2.828427. \quad (4)$$

A dead or short-circuited pipeline could not produce this known nontrivial answer. The vanishing holonomy of the physical transport is therefore a genuine property of the model, not an artifact of the numerics.

F. Oracle versus lattice: the correct labeling

The abstract Ising braid *representation* does contain the non-Abelian structure: the single-exchange commutator $[s_1, s_2]$ has Frobenius norm $\sqrt{2} = \|\sigma_y\|_F$, and the monodromy $[s_1^2, s_2^2]$ has Frobenius norm $2\sqrt{2} = \|2\sigma_y\|_F$. These are properties of the representation — the oracle — obtained by substituting the known F and R data. They answer a *different question* from the dynamical lattice transport, which returns 0. Both numbers are correct, and they must be labeled separately, or a spurious contradiction is invited.

The same discipline applies to the energetics. The oracle’s two-island Ising chain has $E_{\text{GS}} = -(1 + \sqrt{2})$ and gap $\sqrt{2}$, with the $N = 3$ spectrum $\{\pm \cot(\pi/8), \pm 1, \pm \tan(\pi/8)\}$ reproduced at 10^{-15} ; this is a different object from the doubled-Fibonacci lattice of Sec. II, whose ground-state energy is extensive ($E_0 = -4$ on 2×2 , -9 on 3×3). We never mix them.

G. What this section claims, and does not

We claim that on this specific one-dimensionally mapped ladder, the spatial transport of a single anyon provably cannot realize the non-Abelian braid matrix, and that the mechanism — gauge removability together with $\{\mathbb{1}, \psi\}$ superselection — is understood.

We do not claim that non-Abelian braiding is impossible in general, nor that a two-dimensional geometry cannot realize it. And we state plainly that $M_{\tau\tau} = -1/\varphi^2$ of Eq. (1) was the *aim* of the dynamic half and was never measured; this section is the reason why. It must not be confused with the Wilson-loop eigenvalue $+1/\varphi^2$ appearing in the static construction of Sec. II, which is a different quantity with the opposite sign.

V. DISCUSSION AND OUTLOOK

A. The compass

Read together, the two halves point in one direction. Spatial braiding does not live in one-dimensional single-particle transport. That is a transferable lesson rather than a local accident, because both obstructions of Sec. IV C are structural.

The same structural reading explains the static half. The lattice carries the real, basis-invariant modular content because that content is encoded in the operator algebra of the string-net; it carries the chiral sign as well, but through the point-group rotation, not through the Wilson loops, whose real-symmetric holonomies are phase-free. The cut of Sec. III is not where the lattice stops but where the loop route does, and the rotation channel is what crosses it. More broadly, the channel suggests reading modular data off a finite system from a symme-

try of the system at rest, rather than by transporting a particle around it — a perspective we note without claiming it as a demonstrated application.

B. Relation to prior work

We make no priority claim, and one piece of prior work sits close enough that we state it precisely rather than let a reader discover it. Francuz and Dziarmaga [20] extract the topological S and T matrices of non-Abelian string-net models — *including the Fibonacci model treated here* — from an infinite projected entangled pair state, using matrix-product-operator symmetries and their fusion rules to build anyon-flux projectors. The modular data of the Fibonacci string-net are therefore not new with this work, and we do not claim them.

What differs is the method, and with it the question that can be asked. Theirs is a tensor-network computation with exact renormalization-group fixed-point tensors, evaluated in the infinite-cylinder limit; ours is a classically exact diagonalization of a finite lattice, in which a Wilson string is genuinely moved outside the ground space. That difference is not cosmetic here: it is what makes the obstruction of Sec. III A *measurable* — one can ask what a finite real-symmetric holonomy does and does not resolve, and get a number — and it is what makes the transport question of Sec. IV well posed at all. Neither question is posed by an infinite-cylinder ansatz.

Earlier work in the same family determined topological order for the Abelian case [19] and characterized matrix-product-operator symmetries variationally [21]; generalized string-net degeneracies have been analyzed in detail [23]; modular matrices as topological order parameters [24] and their extraction by shearing lattices [25] are established. The rotation channel we use is the construction of Li and Mong [3], whose theorem also bounds what the loop overlaps alone can give; exact diagonalization of a Fibonacci string-net is itself not new, having been used to study the golden-chain transition [4], and minimal quantum circuits for Fibonacci anyons have recently been constructed [5]. Our contribution is the finite-lattice demonstration of the rotation channel, the quantified two-channel error budget, and the measured loop-algebra bound — not the modular data, and not the channel itself.

On the hardware side, Fibonacci statistics have been realized as digital gate sequences [1], and non-Abelian braiding of graph vertices and non-Abelian topological order have been demonstrated on superconducting and trapped-ion processors respectively [26, 27], and Mineev *et al.* [22] prepare a Fibonacci string-net condensate on a superconducting processor and braid its anyons there. These are published positive results and we cite them as such, not as a backdrop of absence. The line to the present work runs the opposite way from the one a reader might expect, so we draw it explicitly: their question is the *physical realization* of the condensate — which

Ref. [22] opens by describing as having remained elusive, and which that work then addresses — whereas ours is a classically exact computation on a finite lattice. The two are complementary, and the present paper contributes nothing to resolving the realization question.

To our knowledge the specific combination reported here — a real-space, out-of-ground-state, chirality-discriminating string in a doubled-Fibonacci Levin–Wen model, the *measured* obstruction of Sec. III A, and the paired no-go of Sec. IV — is independent of that work, but we have not crawled the prior art exhaustively, and we make no claim of being first at anything. Our no-go is independently consistent with the wire-network classification [16, 17], which reaches a compatible conclusion from a different direction. The universality-theoretic backdrop for Fibonacci anyons is due to Freedman, Larsen and Wang [28], with density results due to Kuperberg [29].

C. The remaining niche

One direction from the original program survives, and we place it here rather than in the body because it is off-model. A classically exact, static Hamiltonian, velocity-resolved diabatic-plus-leakage budget for a universal (Fibonacci) anyon would be a genuine target. It belongs to a separate, future strand and not to this paper. We flag it as an open direction, not as a result.

DATA AND CODE AVAILABILITY

The raw JSON outputs and the reproduction script supporting the values reported in this paper are deposited at [doi:10.5281/zenodo.21362245](https://doi.org/10.5281/zenodo.21362245). Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License, Version 2.0 (see LICENSE and LICENSE-CODE).

The deposit contains the load-bearing quantities of Table I together with their provenance. For the static certificate and the no-go: `p6_nogo.json` (the structural no-go route, its two positive controls, and the machinery-alive control), `p6_manifold.json` (the $N = 3$ dense-ED spectrum), `p6_oracle.json` (the algebra oracle, explicitly labeled as such and not as a measurement), `p6_c1_modular.json` (the class-(a)–(c) quantities and the measured loop-route limit, each carrying its evidence class and per-entry provenance), and `p6_modular_reference.py`, a self-contained script (NUMPY only, no engine import, no stored oracle) that rebuilds the analytic doubled-Fibonacci modular data from scratch and checks the analytic targets, including the Gauss sum $p_+ = D$ and the rejection of the order-3 alternative at 5.236. For the rotation channel: `p6_c3_results.json` (the C_3 symmetry and the algebra jump $10 \rightarrow 16$), `p6_lattice_extraction.json` and

`p6.extraction_2x2.json` (the signed S and T extractions on the two sizes), `p6.reconciliation_test.json` and `p6.loop_ideality.json` (the two-channel error budget and the input scale), together with the rotation-channel extractor `p6.extract_c3.py` that produced them, which takes only measured matrices as input, and its pre-registered referee harness `p6.verify_patch.py` (the five acceptance gates and the scaling gate). The error budget is reproducible from this record as well: `p6.reconciliation_test.py` is the deterministic generator of `p6.reconciliation_test.json` and takes only the deposited loop ideality and `p6.extract_c3.py` as input. On both sizes the measured input matrices are deposited, each with its C_3 rotation matrix for the center $(0,0)$: `p6.V_gs`, `p6.Mgs_i`, `p6.Mgs_j` and `p6.M_R` in 3×3 and 2×2 versions, the latter set with the provenance record `p6.manifest_2x2.json`. The extraction therefore runs from this record: applying `p6.extract_c3.py` to the deposited Wilson and rotation matrices, at the deposited loop ideality as input scale and with 240 restarts, returns the four solutions of center $(0,0)$ recorded in `p6.lattice_extraction.json` and the three recorded in `p6.extraction_2x2.json`, reproducing their residuals and sign structure to all printed digits; the README gives both calls verbatim. Both input scales are deposited, not fitted. For 3×3 we additionally deposit one rotation matrix per permutation class, `p6.M_R_3x3_center20.npy` (center $(2,0)$, the second matrix behind the recorded run) and `p6.M_R_3x3_center10.npy` (center $(1,0)$, the class the run did not visit), with `p6.center_independence.py`, which checks from the record that the projected rotation does not depend on the center and that the extractor returns the same block structure and the recorded residuals on each; their construction from the string-net basis is not part of this record. The further 2×2 centers use matrices built from the lattice basis and are not deposited. Note that the Wilson matrix carrying the extraction is `p6.Mgs_j_3x3.npy` on one size and `p6.Mgs_i_2x2.npy` on the other: the convention does not carry over, and the README states why. The residuals are not an output of the extractor. They are formed in a separate scoring step against the analytic doubled-Fibonacci target that `p6.modular_reference.py` rebuilds from scratch; the driver performing that step is released with this record, while the generation of the measured matrices themselves is not, since the exact-diagonalization engine that produces the ground-state and loop matrices is a separate, heavier computation. A minimal ED path (`p6.ed_path.py`) and its geometry record (`p6.geometry_2x2.json`) identify the row ordering of the deposited basis; that path is not the production engine.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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